Lab: AWS Services I

DS 7200 Distributed Computing — Basics of AWS Identity and Access Management (IAM)

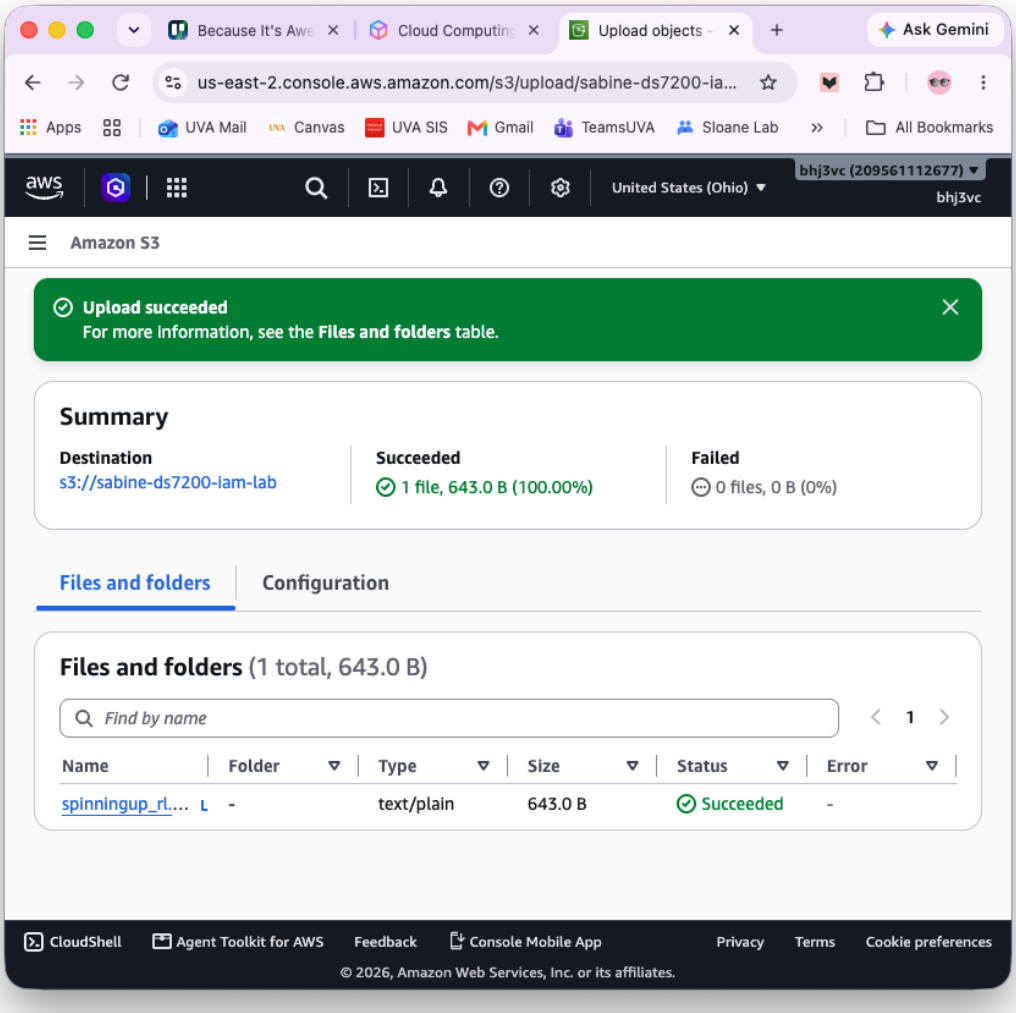
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Computing ID: bhj3vc

Date: November 2, 2026

1. Create an S3 bucket and upload spinningup_rl.txt (1 point)

Created bucket sabine-ds7200-iam-lab (us-east-2) and uploaded spinningup_rl.txt.



2. S3 URI (1 point)

`s3://sabine-ds7200-iam-lab/spinningup_rl.txt`

3. Create a role in IAM

Started creating the role data_scientist_s3_role in IAM > Roles > Create role.

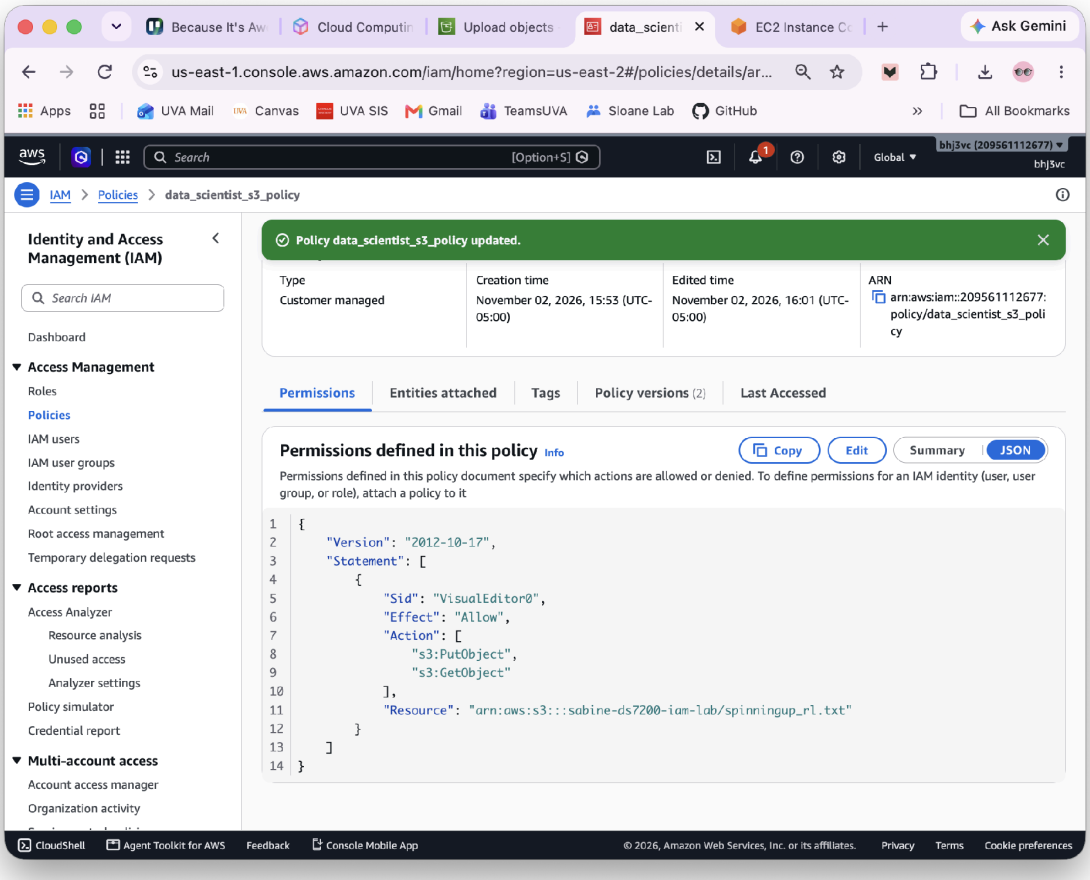
4. Allow common use case: EC2

Trusted entity type: AWS service; use case: EC2.

5. Create a policy with S3 GetObject and PutObject

Created a customer-managed policy allowing s3:GetObject and s3:PutObject on the single object: `arn:aws:s3:::sabine-ds7200-iam-lab/spinningup_rl.txt`

6. Policy summary: data_scientist_s3_policy (1 point)



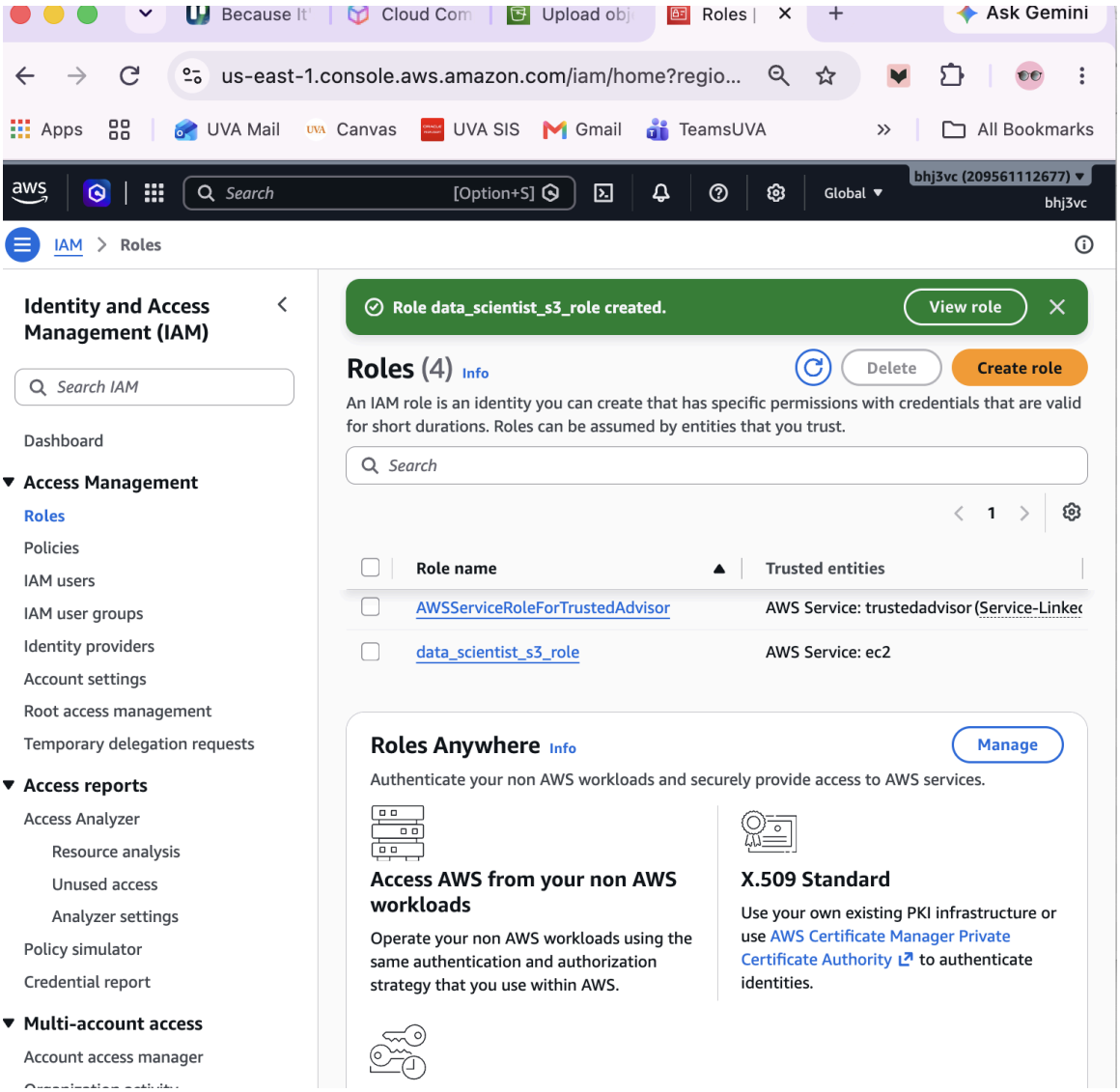
7. Attach the policy to the role

Selected data_scientist_s3_policy on the role's Add permissions page.

8. Final role creation step

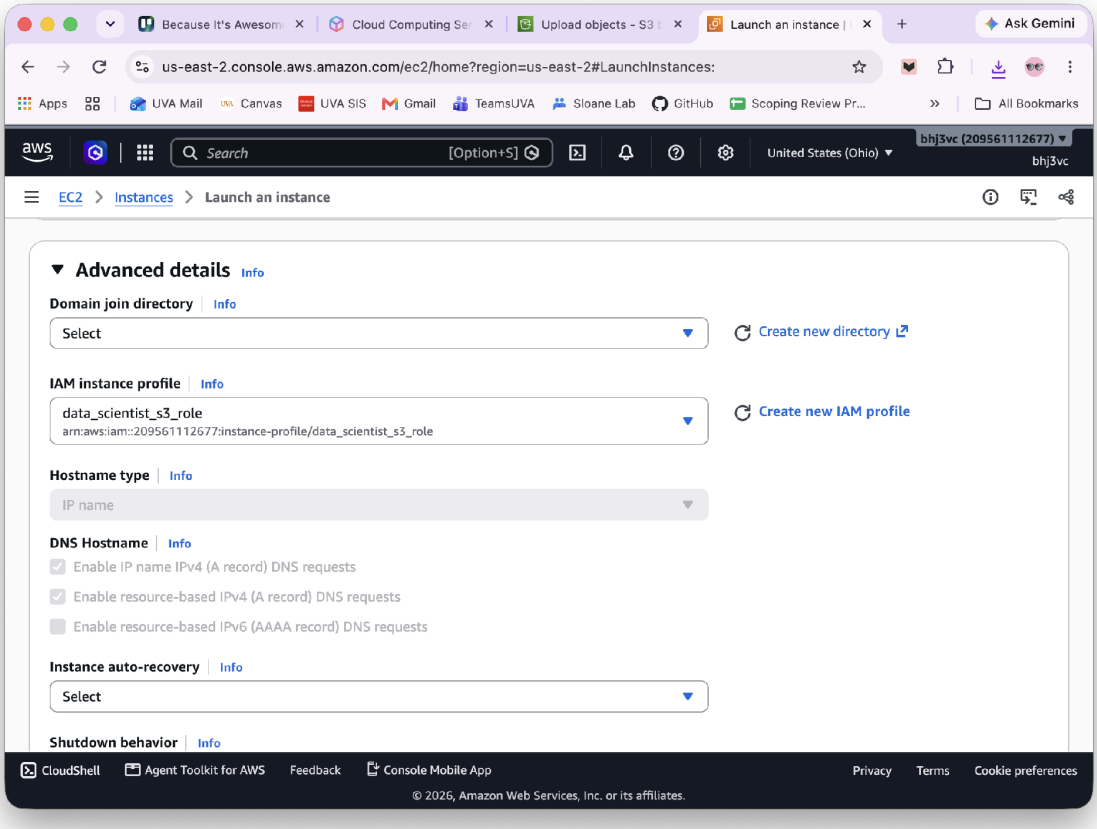
Named the role data_scientist_s3_role; the trust policy lists ec2.amazonaws.com as the trusted service.

9. Roles page showing the new role (1 point)



10. Create an EC2 instance (1 point)

t3.micro, Amazon Linux 2023, new key pair, SSH and HTTPS allowed. IAM instance profile set to data_scientist_s3_role.



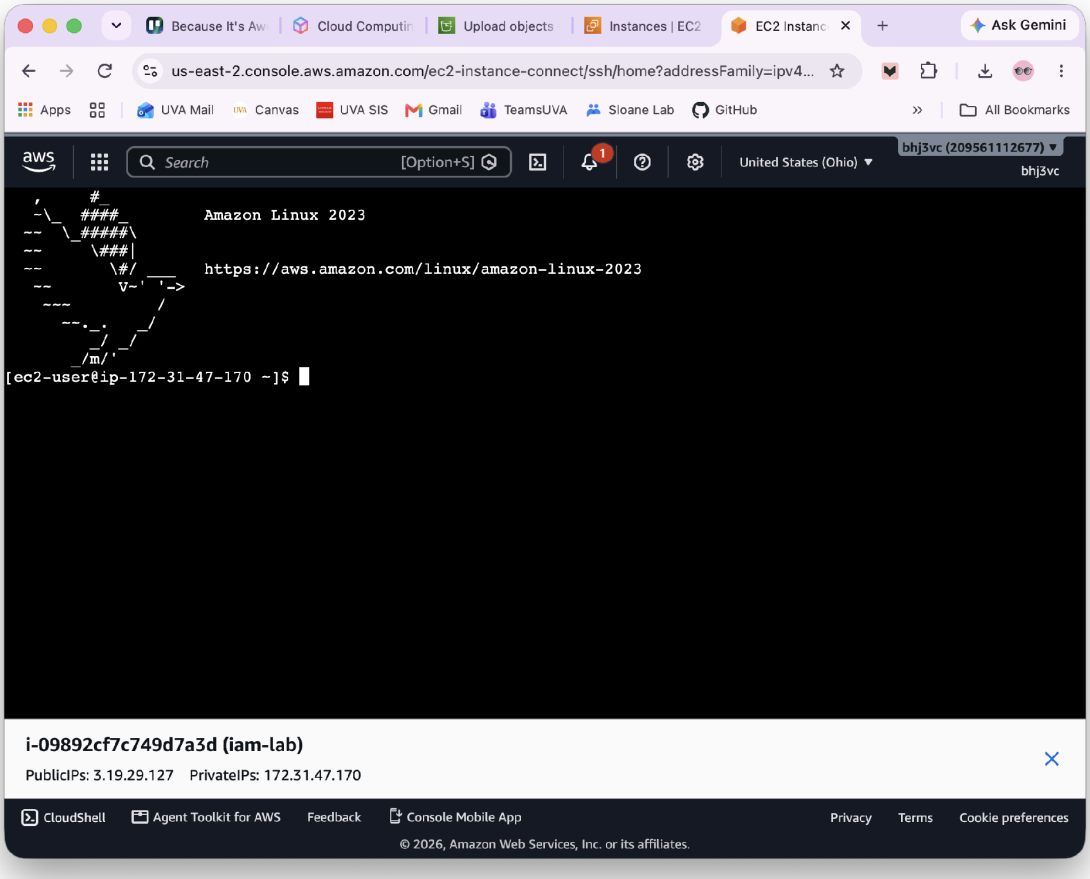
11. Launch the instance

Launched instance iam-lab (i-09892cf7c749d7a3d).

12. Connect to the instance

Connected with EC2 Instance Connect as ec2-user.

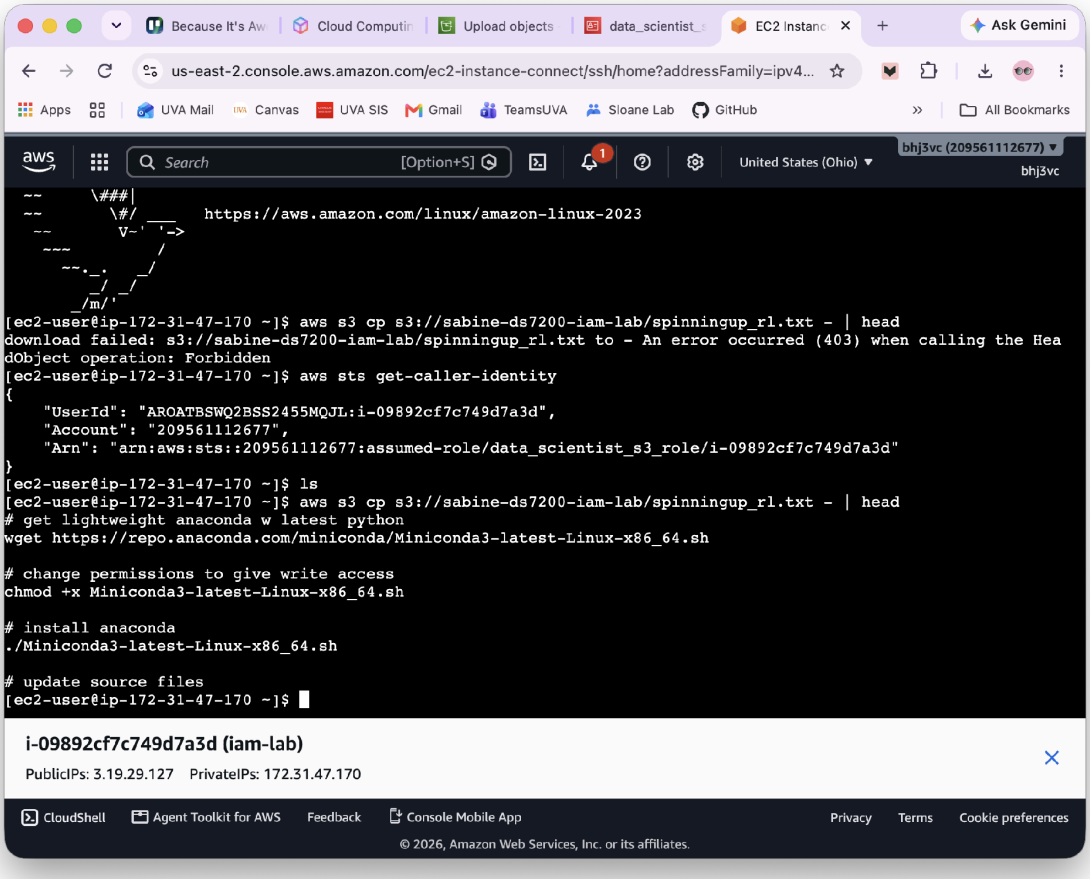
13. EC2 landing page (1 point)



14. Read the S3 file from EC2 (1 point)

```
aws s3 cp s3://sabine-ds7200-iam-lab/spinningup_rl.txt - | head
```

The first attempt returned 403 Forbidden because the policy's resource ARN had been entered incorrectly. After correcting the ARN in step 5's policy, the same command printed the top lines of the file.



15. Install Miniconda

```
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh
```

```
chmod +x Miniconda3-latest-Linux-x86_64.sh
```

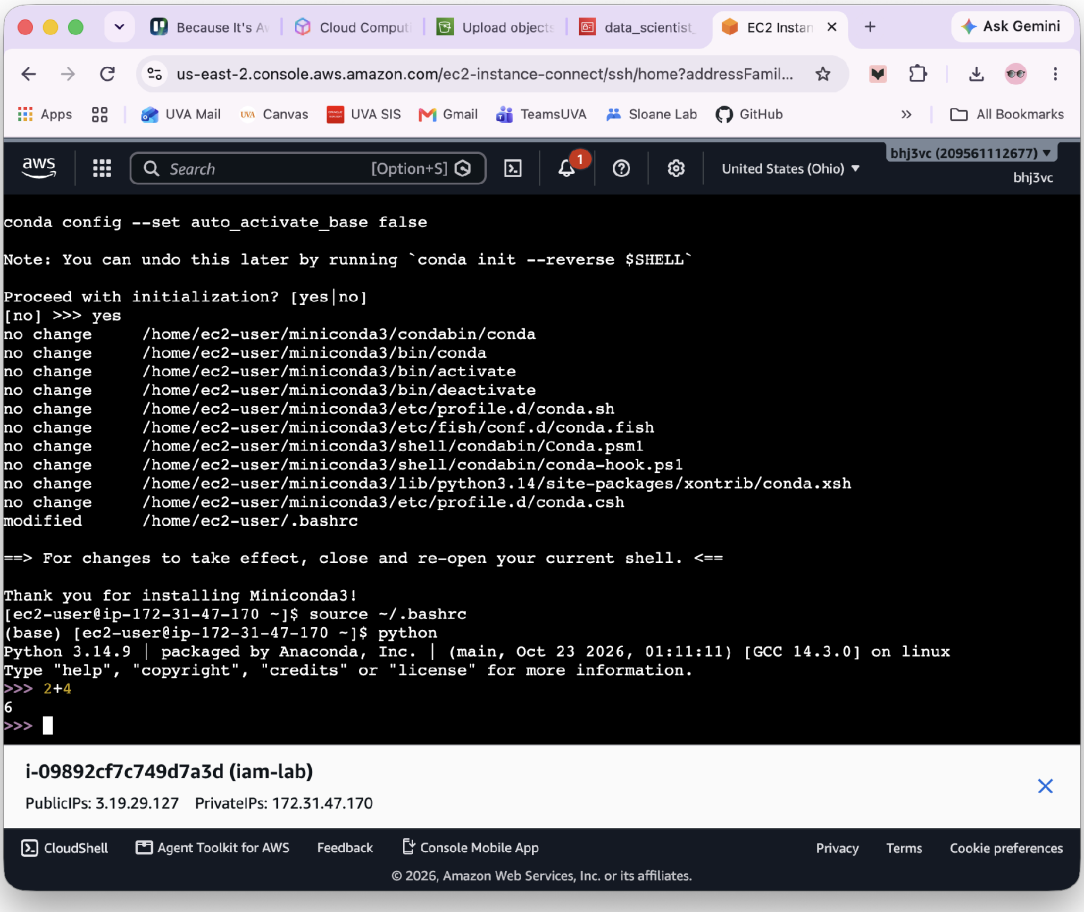
```
./Miniconda3-latest-Linux-x86_64.sh
```

16. Update the session

```
source ~/.bashrc
```

17. Python is installed (1 point)

Python 3.14.9 (installed by the latest Miniconda).



18. Terminate the EC2 instance (1 point)

